

Phytochemicals for breast cancer prevention by targeting aromatase.

Aromatase is a cytochrome P450 enzyme (CYP19) and is the rate limiting enzyme in the conversion of androgens to estrogens. Suppression of in situ estrogen production through aromatase inhibition is the current treatment strategy for hormone-responsive breast cancers. Drugs that inhibit aromatase have been developed and are currently utilized as adjuvant therapy for breast cancer in post-menopausal women with hormone dependent breast cancer. Natural compounds have been studied extensively for important biologic effects such as antioxidant, anti-tumor and anti-viral effects. A significant number of studies have also investigated the aromatase inhibitory properties of a variety of plant extracts and phytochemicals. The identification of natural compounds that inhibit aromatase could be useful both from a chemopreventive standpoint and in the development of new aromatase inhibitory drugs. This review will discuss whole food extracts and the common classes of phytochemicals which have been investigated for potential aromatase inhibitory activity. We will review reported aromatase inhibition, kinetic data and possible structural variations that may inhibit or enhance the interaction of phytochemicals with the aromatase enzyme.